

Colorado River Basin Regional Water Quality Control Board

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December 13, 2019

Michael Hulsey
Calexico Border Patrol Station
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**SUBJECT: NOTICE OF APPLICABILITY FOR ENROLLMENT IN GENERAL
ORDER WQ-2014-0153-DWQ**

**FACILITY: CALEXICO BORDER PATROL STATION WASTEWATER TREATMENT
PLANT, CALEXICO, IMPERIAL COUNTY**

Dear Mr. Hulsey:

The California Regional Water Board Quality Control Board, Colorado River Basin Region (Regional Water Board) has reviewed the Report of Waste Discharge (ROWD) and Engineering Report dated July 9, 2019, submitted by the U.S. Customs and Border Protection (Discharger) for the Calexico Border Patrol Station Wastewater Treatment Plant (WWTP or Facility).

This letter serves as a Notice of Applicability (NOA) to enroll the Calexico Border Patrol Station in Order WQ 2014-0153-DWQ, *General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* (General Order). The Facility is assigned Waste Discharger Identification (WDID) 7A130139001, and GeoTracker Global ID WDR100045826.

A copy of the General Order is available online at:

https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wq2014_0153_dwq.pdf.

You must familiarize yourself with the contents of the General Order. You must manage the discharge in accordance with the requirements contained in the General Order, the information submitted in the ROWD and supplemental correspondence, the requirements contained in this revised NOA, and the requirements of Monitoring and Reporting Program (MRP) WQ 2014-0153-DWQ(9).

FACILITY AND DISCHARGE DESCRIPTION

The Facility is located at 536 Barbara Worth Road, Calexico, California 92231, approximately one mile east of the Calexico city limits, near the intersection of State Route 98 and Barbara Worth Road in Imperial County. The Facility is operated by the United States Border Patrol (USBP), a component within Customs and Border Protection. The current wastewater disposal method on site is tank and haul, where wastewater is stored in septic tanks and are hauled away to a wastewater treatment plant. The Assessor's Parcel Number is 059-190-19. The Township and Range are a portion of Tract 61, Section 9, Township 17 South, Range 15 East San Bernardino Meridian. The Latitude is 32.689499 °N and the Longitude is 115.430999 °W.

The current wastewater disposal method is outdated and inadequate to support operational requirements at the Facility. The Discharger plans to expand the onsite wastewater treatment system (OWTS) with an advanced onsite wastewater treatment facility and an evaporation pond. The new system will be capable of treating 5,000 gallons per day (gpd) of primarily domestic wastewater generated by employees and detainees at the Facility. Minor wastewater flows of 300 gpd to 500 gpd include membrane backwash from potable water treatment system consisting of a membrane filtration system, ultraviolet disinfection units and hypochlorite injection. The waste stream will also include 300 gpd of vehicle wash water, which is treated by an oil/dust separator prior to discharging to a septic tank. The biofiltration will be installed north of the Administration Building, east of the Alamitos Canal as shown on **Attachment 1**. The Facility is located in a remote area with no access to municipal water or wastewater systems as shown in **Attachment 2**. The property is relatively flat, at an elevation of 15 to 20 feet above mean sea level (MSL). The site is in an area of minimal flood hazard, Zone X, and not subject to a 100-year flood event, as defined by FEMA.

The expansion to the OWTS will use a biofiltration system that uses earthworms and microbial processes to provide secondary treatment to the wastewater. The system consists of a bed of wood shavings underlain by drain rock and an effluent recovery system. The biofiltration system uses an industry-specific mix of worms and bacteria to achieve maximum reduction efficiencies on parameters such as Biochemical Oxygen Demand (BOD); total suspended solids (TSS); fats, oils, and grease (FOG); total Nitrogen; ammonia; and phosphorous. The burrowing worms create air channels, digest suspended solids, and can achieve densities of 12,000 worms per cubic yard. The worm beds are not in direct sunlight and include a solar cooling fan. Wastewater is screened and adjusted for pH before being intermittently sprinkled over the surface of the bed. The drain rock layer has passive vents.

The process, shown in **Attachment 3**, is similar to a recirculating gravel filter. The recirculating gravel filter consists of a sprinkler system of intermittent application to a bed of pea gravel underlain by drain rock. The hydraulic loading rates for recirculating gravel filters range from 3 to 5 gallons per day per square foot (gpd/ft²). The BOD loading rates for recirculating gravel filters range from 87 to 348 lb/acre-day, which is similar to low-rate trickling filters.

The treated wastewater will be conveyed to an adjacent evaporation pond. The pond has a water surface area of 1 acre and a capacity of 512,000 gallons, at 2 feet deep, with 2 feet of freeboard. The existing septic tank will remain onsite and provide primary solids removal prior to treatment in the biofiltration unit. Sludge and solid waste from the treatment facility will be periodically hauled away. The Water Balance calculation performed shows that the size of the pond is sufficient to accommodate a daily loading of 5,000 gpd. Best practices will be observed to minimize objectionable odors, minimize weeds and algae growth, and prevent breeding of mosquitos.

REGULATORY CONSIDERATIONS

With respect to groundwater quality objectives, the Water Quality Control Plan for the Colorado River Basin Region (Basin Plan) states that the Regional Water Board's goal is to maintain the existing water quality of all non-degraded groundwater basins. However, in most cases ground water pumped generally returns to the basin after use with an increase in mineral concentrations, such as total dissolved solids (TDS), nitrates, etc. Under these circumstances, the Regional Water Board's objective is to minimize the quantities of contaminants reaching any groundwater basin. Groundwater in the area of the discharge has a municipal (MUN) beneficial use designation. The Basin Plan further states that ground waters designated for use as domestic or municipal supply shall not contain concentrations of chemical constituents in excess of the limits specified in California Code of Regulations, title 22, chapter 15, article 4, section 64435, Tables 2, 3, and 4 as a result of human activity. While it is anticipated that some degradation of groundwater quality may occur due to the discharge, the increase in the concentrations of the constituents of concern is not expected to be significant. In addition, due to the low volume of the discharge, water quality objectives will not be violated.

CEQA EXEMPTION

The Regional Water Board has determined that the activities authorized by this NOA are exempt from review under the California Environmental Quality Act (CEQA) pursuant to California Code of Regulations, title 14, section 15061. Specifically, the activities authorized by this NOA meet the exemption criteria under California Code of Regulations, title 14, section 15301 (Existing Facilities), section 15302 (Replacement or Reconstruction), section 15303 (New Construction or Conversion of Small Facilities), and/or section 15311 (Accessory Structures). There is already an existing OWTS, which the Discharger plans on expanding with a new vermifiltration wastewater treatment system. There is no substantial evidence that any exceptions to the foregoing CEQA categorical exemptions apply. Additionally, the State Water Resource Control Board (State Water Board) considered the potential environmental impacts associated with new or expanding small domestic systems enrolled in the General Order in a Negative Declaration adopted on September 23, 2014 through State Water Board Resolution 2014-0054.

SITE SPECIFIC REQUIREMENTS

The operator of the WWTP shall be familiar with the requirements contained in the General Order, this NOA, and the Monitoring and Reporting Program (MRP).

The General Order contains prohibitions and specifications that apply to all wastewater systems, and specifications that apply only to specific treatment and/or disposal systems. The Discharger shall comply with the following specific requirements:

- a. Required Technical Reports/Plans--By **February 14, 2020**, the Discharger shall submit to the Regional Water Board the following plans:
 - A Spill Prevention and Emergency Response Plan consistent with the requirements of General Order Provision E.1.a; and
 - A Sampling and Analysis Plan consistent with the requirements of General Order Provision E.1.b

MONITORING AND REPORTING

The Discharger shall comply with the requirements of MRP WQ 2014-0153-DWQ(9), as provided in this section. The MRP is effective upon issuance of this NOA. The MRP requires the Discharger to monitor the WWTP, collect and analyze samples according to the following schedule, and report results quarterly and annually.

Water Supply Influent Monitoring

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
TDS	mg/L	Grab	Quarterly	Quarterly

Septic Tank Influent Monitoring

Influent samples shall be taken from a location that provides representative samples of the wastewater quality discharged into the septic tank. At a minimum, septic tank influent monitoring shall consist of the following:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Total Nitrogen	mg/L ¹	Grab	Annually	Annually
BOD	mg/L	Grab	Annually	Annually
TSS	mg/L	Grab	Annually	Annually

Biofiltration Influent Monitoring

Influent samples shall be taken from a location that provides representative samples of the wastewater quality discharged into the biofiltration unit. At a minimum, biofiltration unit influent monitoring shall consist of the following:

<u>Constituent</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Total Nitrogen	mg/L	Grab	Quarterly	Quarterly
BOD	mg/L	Grab	Quarterly	Quarterly
TSS	mg/L	Grab	Quarterly	Quarterly

Biofiltration Effluent Monitoring

Samples of effluent shall be taken at an area that represents the biofiltration effluent quality distributed to the disposal area, at a minimum, biofiltration effluent monitoring shall consist of the following:

<u>Parameters</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Flow Rate	gpd	Metered ²	Continuous	Quarterly
BOD	mg/L	Grab	Quarterly	Quarterly
TSS	mg/L	Grab	Quarterly	Quarterly
Total Nitrogen	mg/L	Grab	Quarterly	Quarterly
pH	s.u. ³	Grab	Quarterly	Quarterly
TDS	mg/L	Grab	Quarterly	Quarterly

¹ mg/L milligrams per liter.

² Flow rate may be metered or estimated based on potable water supply meter readings or other approved method. Flow rates may be measured as influent or effluent flow.

³ Standard pH units

Evaporation Pond Monitoring

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sample Frequency</u>	<u>Reporting Frequency</u>
Free Board	0.1 feet	Measurement	Quarterly	Quarterly
Odors	--	Observation	Quarterly	Quarterly
Berm Condition	mg/L	Grab	Quarterly	Quarterly

FACILITY REQUIREMENTS AND DISCHARGE FLOW LIMITATION

1. The Discharger shall comply with General Order WQ 2014-0153-DWQ.
2. The flow of the wastewater discharged to the evaporation pond shall not exceed a daily maximum of 5,000 gallons.
3. Wastewater discharged to the evaporation pond shall comply with the following effluent limits:

Constituent	units	30-day average	7-day average
BOD	mg/L	30	45
TSS	mg/L	30	45

4. **Applicable Requirements of the General Order (Septic Systems).** The Discharger must comply with Section B.2 (Septic Systems) of the General Order, which includes requirements such as preventing the disposal of household hazardous waste and toxic chemicals, minimizing the disposal of brines, phosphates and other inorganic salts to the OWTS; and discouraging the use of holding tank additives/chemicals that may contain formaldehyde, zinc, phenol, and other chemicals (see Attachment B2 of the General Order).

WATER CODE SECTION 13267 ORDER FOR TECHNICAL REPORTS

In order to ensure compliance with the specified design and compliance with the conditions of the General Order, the Discharger is hereby directed pursuant to section 13267 of the California Water Code to submit the information described below.

1. **Within 60-days** of issuance of this NOA, the Discharger shall submit a schedule for construction of the onsite wastewater treatment and disposal system.
2. **No later than 3 days** prior to the start of construction of the onsite wastewater treatment and disposal system, the Discharger shall submit copies of all local permits obtained for that purpose, and notify the Regional Water Board verbally and via email for the purpose of establishing appointments for inspections.

3. **Within 30-days** of project completion, the Discharger shall submit a copy of the contract with the operator contracted for the maintenance and operation of the WWTF. The documentation shall include the detail necessary to demonstrate compliance to the General Order, including solids management and cite contact information and required operator certification.
- 4 **Within 6 months** of project completion, the Discharger shall submit as-built documents depicting the wastewater treatment and disposal evaporation pond.
- 5 **Within 3 months** prior to changes to the character of the wastewater or method or location of treated and disposal from that described in the ROWD, the Discharger must submit a new ROWD, or notification that the facility is connecting to a centralized sewer system.
- 6 **The Discharger shall pay** the required annual fee specified in the annual billing from the State Water Board until this NOA is officially terminated. The Discharger must notify the Regional Water Board in writing if the discharge regulated by this NOA ceases so that we may terminate coverage and avoid unnecessary billing.

In accordance with California Business and Professions Code Sections 6735, 7835, and 7835.1, engineering and geologic evaluations and judgments shall be performed by or under the direction of California registered professionals (i.e., civil engineer, engineering geologist, geologist, etc.) competent and proficient in the fields pertinent to the required activities. All technical reports that contain work plans, describe the conduct of investigations and studies, or that contain technical conclusions and recommendations concerning engineering and geology shall be prepared by or under the direction of appropriately qualified professional(s), even if not explicitly stated. The report must contain a statement of qualifications of the licensed professional(s) in responsible charge of this report, as well as the professional's signature and stamp.

All technical reports shall include a statement by the Discharger, or an authorized representative of the Discharger, certifying under penalty of perjury under the laws of the state of California, that the reports were prepared under his or her supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluated the information submitted, and that based on his or her inquiry of the person or persons who manage the system, the information submitted is, to the best of his or her knowledge and belief, true, complete, and accurate.

OTHER REQUIREMENTS

Please review this NOA carefully to ensure that it completely and accurately reflects the proposed discharge. If the discharge violates the terms or conditions of the General Order or this NOA, the Regional Water Board may take enforcement action, including assessment of administrative civil liability of up to \$1,000 per day for late or inadequate reports. If the method or location of treated domestic wastewater disposal changes from

that described in the ROWD, you must submit a new ROWD or Notice of Intent (NOI) to comply with the conditions of General Order WQ 2014-0153-DWQ.

Documents that are normally sent via mail by the Discharger, such as regulatory reports, documents, submissions, materials, data, and correspondence, to the Regional Water Board shall be converted to Portable Document Format (PDF) or other appropriate Microsoft application, such as Excel, and submit to the State Water Board GeoTracker database found at: <https://geotracker.waterboards.ca.gov/>. Documents that are too large (400 MB or larger) should be broken down into smaller electronic files, labelled properly and uploaded into GeoTracker. In the subject line, when submitting monitoring reports and documents, please reference General Order number WQ 2014-0153-DWQ(9).

Should you have any questions concerning this NOA, please contact Reginald Tan, Water Resource Control Engineer, at (760) 776-8944, reginald.tan@waterboard.ca.gov or Jose Cortez, Land Disposal Unit Chief, at (760) 776-8963 or at jose.cortez@waterboards.ca.gov.

Sincerely,



Paula Rasmussen
Executive Officer
Colorado River Basin
Regional Water Quality Control Board

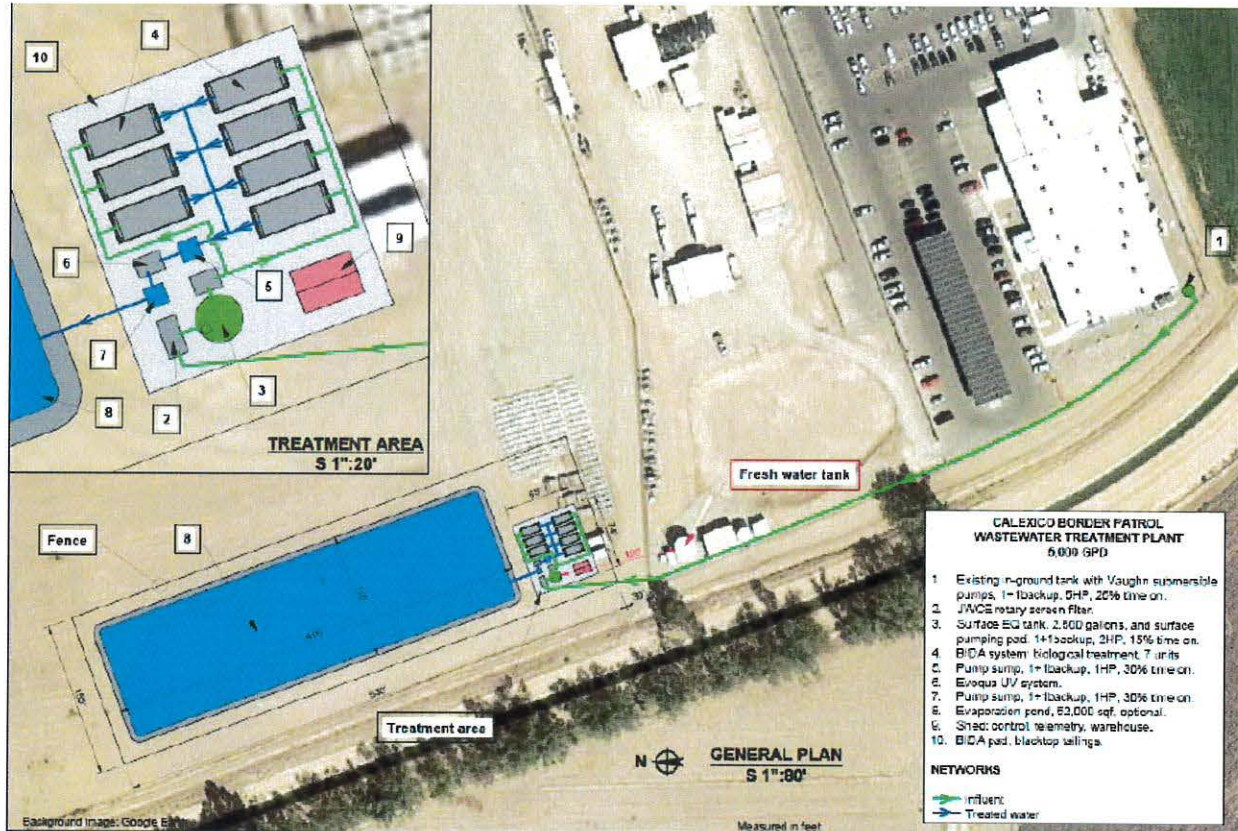
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Enclosures: General Order WQ 2014-0153-DWQ

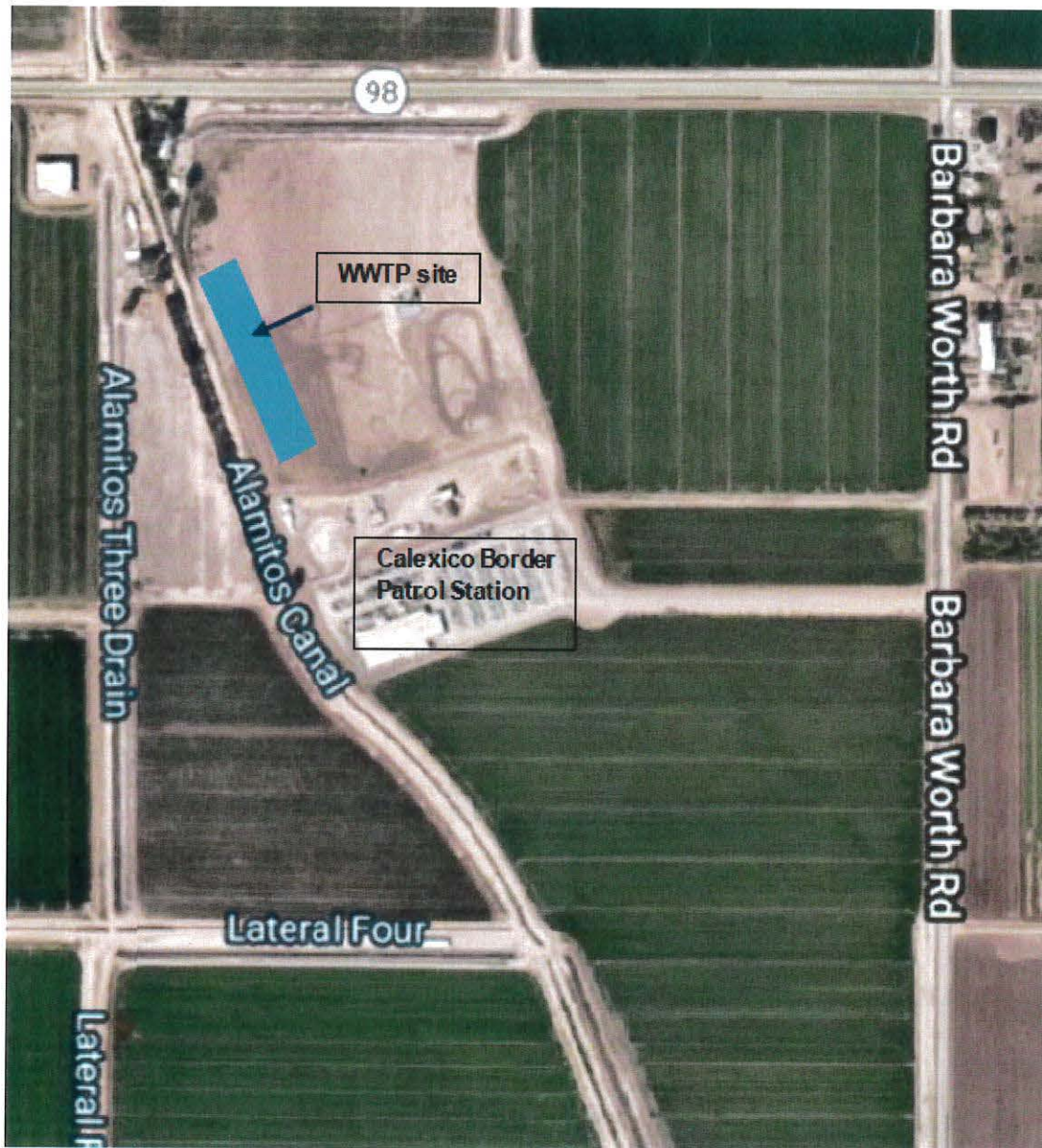
cc: Jennifer Duffy, P.E., HDR Inc., Jennifer.duffy@hdrinc.com

File: 7A130139001, Calexico Border Patrol Station WWTP, WQ 2014-0153-DWQ(9)

Attachment 1: Site Map



Attachment 2: Vicinity Map



Attachment 3: Wastewater Treatment Plant Process

